METRIC

A-A-59106B
27 January 2020
SUPERSEDING
A-A-59106A
04 August 2014

COMMERCIAL ITEM DESCRIPTION

ETHYLENE GLYCOL MONOETHYL ETHER, TECHNICAL

The General Services Administration has authorized the use of this commercial item for all federal agencies.

1. SCOPE. This commercial item description covers technical grade ethylene glycol monoethyl ether, herein referred to as the solvent.

2. SALIENT CHARACTERISTICS

2.1 Appearance. The solvent shall be clear and uniform when visually inspected using transmitted light.

2.2 Chemical and physical characteristics. The solvent shall conform to the chemical and physical characteristics of Table I when tested as specified therein.

2.3 Spot test. Support a piece of 125 millimeter (mm) diameter filter paper (Whatman No. 2 or equal) on a 70 mm diameter crystallizing dish or equivalent. Using a small pipette or burette, place five drops of the solvent on the center of the paper. Allow the liquid to evaporate at room temperature (20-25 ºC), away from both sunlight and strong drafts. After 30 minutes examine the filter paper for the presence of any stains or residual oiliness.

2.4 Relative evaporation rate test. A set volume of the solvent shall be placed in a controlled environment (such as a thermogravimetric analyzer) with an equilibrium temperature of 25 ± 0.5°C, a humidity less than 5%, and a gas flow rate of 21 ± 1 L/min. The change in weight shall be recorded throughout the test until at least 90% of the solvent has evaporated. The same volume of n-buytl acetate shall be tested under the same conditions and the relative evaporation

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rate shall be determined as the ratio of the 90% evaporation times. The volume of solvent, instrument used, and any other testing parameters shall be selected to ensure duplicate results on separate days vary by less than 5.1% at the 95 % confidence level. A volume of 0.7 mL has proven sufficient for achieving this repeatability when using a Shell thin-film evaporometer according to manufacturer recommendations.

TABLE I. Chemical and physical characteristics.

Characteristics	Requirements		Test Method
	Min	Max	
Color, Pt-Co Scale	---	Number 15	ASTM D1209
Odor	---	Mild, Non-residual	ASTM D1296
Water content, percent by weight	---	0.1	ASTM D1364
Specific gravity at 20º C /20º C	0.929	0.932	ASTM D891(1)
Nonvolatile matter (mg/100 ml sample)	---	5	ASTM D1353
Acidity (free acid as acetic acid), percent by weight	---	0.01	ASTM D1613
Distillation range, ºC	134.0	136.0	ASTM D1078(2)
Flash Point, ºC	38.0	---	ASTM D56
Spot test	---	No stains or residual oiliness	See 2.3
Relative evaporation rate (n-butyl acetate = 1.0)	---	0.34	See 2.4
Sulfur	---	Negative	ASTM D4952

(1) In case of a dispute, the pycnometer method shall be used as the referee method.

(2) Using ASTM Solvents Distillation Thermometer 102C, with a range from 123°C to 177°C.

3. REGULATORY REQUIREMENTS

3.1 Labeling, packaging and marking. Item shall be labeled, packed and marked in accordance with Title 49 Code of Federal Regulations (CFR), Parts 171 to 199.

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3.2 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this CID are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this CID.

3.3 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations § 1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance. Any certificate of analysis (COA) offered with the product to the commercial market shall be provided with each lot or batch of product. The lot or batch number and the date of inspection/analysis from the offer shall be on each COA.

4.2 Market acceptability. The item offered must have been sold to the government or commercial market.

5. PACKAGING. Preservation, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Electronic copies of ASTM document are available online at http://www.astm.org/.

6.1.2 Electronic copies of CFR documents can be obtained from https://gov.ecfr.io/cgi-bin/ECFR.

6.1.3 Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/browse/index/far.

6.2 Ordering data. Acquisition documents must specify the following:

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a. Title, name, and date of this CID.
b. Unit quantity required.
c. Packaging requirements.

6.3 National stock numbers (NSNs). The following NSNs correspond to this CID. This list may not be indicative of all possible NSNs associated with this document.

6810-00-222-2751
6810-00-285-4309

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.5 Key Words.

2-ethoxyethanol
Cellosolve
Multi-purpose cleaner
Varnish remover

6.6 Changes from previous issues. Marginal notations are used in this revision to identify changes with respect to the previous issue.

MILITARY INTERESTS:

Custodians:
Army - EA
Air Force - 68
DLA - GS

Reviewers:
Army - MD1

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FAS

Preparing Activity:
DLA – GS3
(Project No: 6810-2020-001)

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